Game-informed MCDA decision-support report v0.9.0

Project: Primary care funding architecture in Australia and New Zealand
Focus: New Zealand upstream access, primary care, ambulance and hospital-pressure games
Version: v0.9.0
Status: Demonstrative decision-support layer; not empirically calibrated

Executive summary

This report adds a game-informed MCDA layer to the final hybrid model. The purpose is to help decision-makers evaluate whether the mapped games matter, how strongly each game contributes to the policy problem, and which policy options are most likely to move the system toward better equilibria.

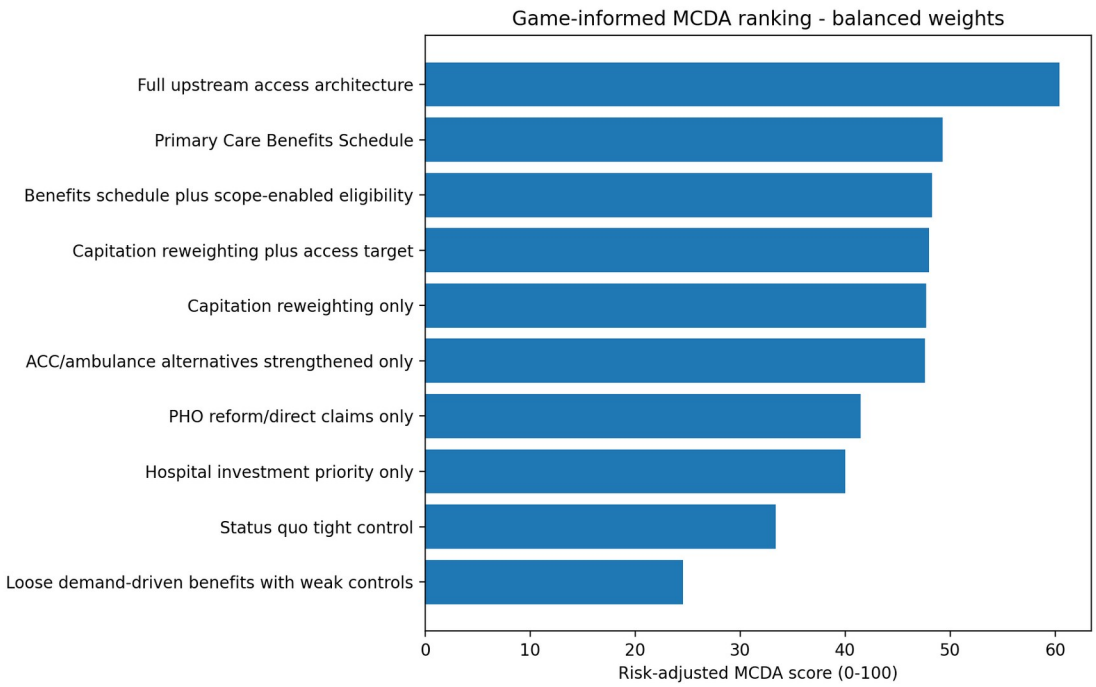
The MCDA does not replace empirical modelling. It makes judgement explicit. It is designed for RACMA discussion, stakeholder workshops, policy option screening and validation planning.

The central thesis remains:

> New Zealand may be managing primary care and ambulance activity so tightly that unmet need is channelled into hospitals by default.

The MCDA translates that thesis into two decision-support tools:

1. **Diagnostic game-position mapping**: where does each game currently sit, and how important is it?
2. **Policy-option MCDA**: which reform option best balances access, equity, hospital deflection, fiscal control, governance, market entry and feasibility?



Default MCDA ranking

Why MCDA is useful here

This policy problem is not just a technical funding-design problem. It also involves values, risk tolerance, professional boundaries, equity, Te Tiriti legitimacy, rural resilience, fiscal exposure, data observability and political feasibility.

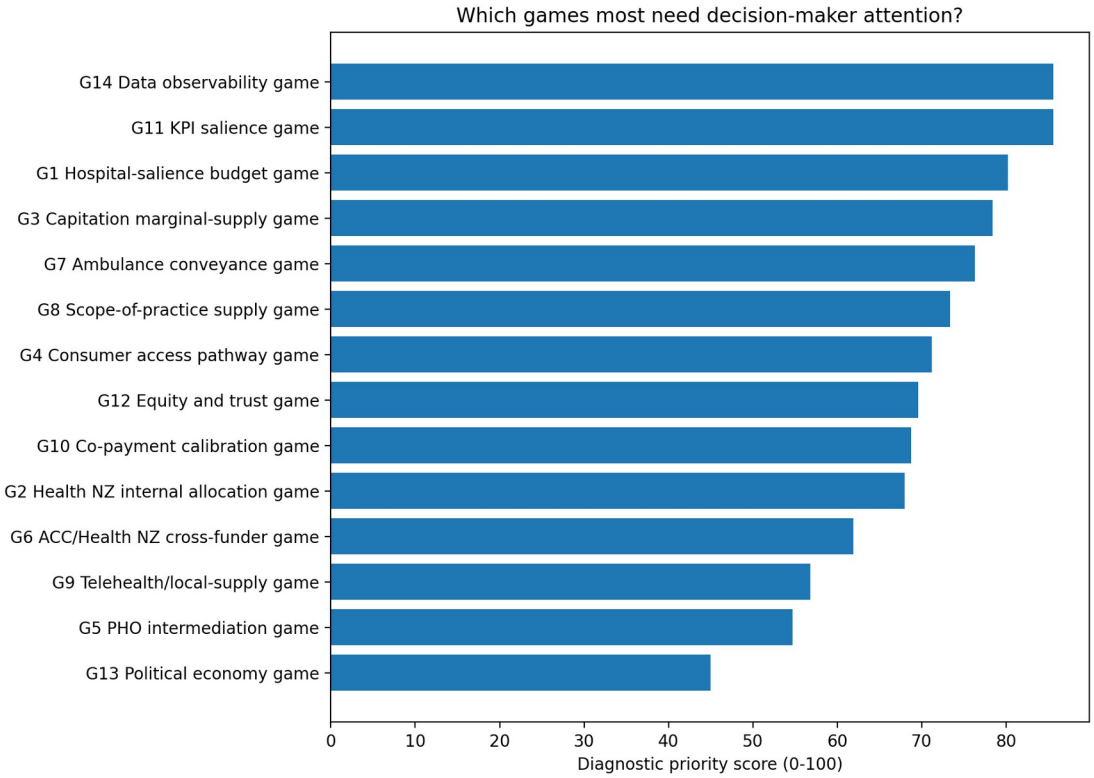
A pure model can show what follows from assumptions. MCDA shows which assumptions and values decision-makers are using.

Diagnostic game map

The highest-priority diagnostic games in the example scoring are:

game_id	game_name	current_equilibrium	better_equilibrium	priority_score_0_to_100
G11	KPI salience game	hospital-target dominance	upstream target salience	85.6
G14	Data observability game	hidden-unmet-need equilibrium	observable upstream-flow equilibrium	85.6
G1	Hospital-salience budget game	hospital-rescue equilibrium	upstream-salience equilibrium	80.2
G3	Capitation marginal-supply game	marginal-rationing equilibrium	marginal-expansion equilibrium	78.4

game_id	game_name	current_equilibrium	better_equilibrium	priority_score_0_to_100
G7	Ambulance conveyance game	ED-conveyance default	safe alternative-disposition equilibrium	76.3
G8	Scope-of-practice supply game	professional-bottleneck equilibrium	scope-enabled supply equilibrium	73.4
G4	Consumer access pathway game	delay/pay/ED substitution	early-access equilibrium	71.2
G12	Equity and trust game	transactional-access without trust	benefits plus equity-function equilibrium	69.6



Diagnostic game priority

The diagnostic map suggests that KPI salience, data observability, hospital salience, capitation marginal supply, ambulance conveyance and scope-of-practice supply are especially important games to test and discuss.

Policy-option MCDA

The default MCDA criteria are:

criterion_id	criterion	default_weight	related_games
C1	Access and supply generation	14	G3, G4, G8

criterion_id	criterion	default_weight	related_games
C2	Hospital deflection	14	G1, G2, G7
C3	Equity and Te Tiriti legitimacy	14	G10, G12
C4	Rural and in-person resilience	9	G7, G8, G9
C5	Fiscal sustainability	12	G1, G2, G6, G10
C6	Gaming and low-value activity risk	10	G3, G5, G10, G13
C7	Administrative simplicity and market entry	9	G5, G8, G14
C8	Governance and clinical safety	10	G8, G11, G14
C9	Political feasibility	5	G5, G12, G13
C10	Data and accountability readiness	8	G11, G14

The default policy options are:

option_id	option	mapped_scenario
O0	Status quo tight control	S0
O1	Capitation reweighting only	S1
O2	Capitation reweighting plus access target	S1+target
O3	Primary Care Benefits Schedule	S2
O4	Benefits schedule plus scope-enabled eligibility	S2+scope
O5	Full upstream access architecture	S3
O6	Loose demand-driven benefits with weak controls	S4
O7	ACC/ambulance alternatives strengthened only	partial
O8	PHO reform/direct claims only	partial
O9	Hospital investment priority only	hospital

Balanced-weight result

rank	option_id	option	weighted_total_before_penalty	risk_penalty	risk_adjusted_score
1	O5	Full upstream access architecture	70.77	10.35	60.42
2	O3	Primary Care Benefits Schedule	59.37	10.1	49.27
3	O4	Benefits schedule plus scope-enabled eligibility	60.58	12.3	48.28
4	O2	Capitation reweighting plus access target	54.5	6.5	48
5	O1	Capitation reweighting only	52.89	5.2	47.69
6	O7	ACC/ambulance alternatives strengthened only	55.78	8.2	47.58

rank	option_id	option	weighted_total_before_penalty	risk_penalty	risk_adjusted_score
7	O8	PHO reform/direct claims only	53.72	12.25	41.47
8	O9	Hospital investment priority only	48.04	8	40.04
9	O0	Status quo tight control	40.9	7.5	33.4
10	O6	Loose demand-driven benefits with weak controls	47.33	22.75	24.58

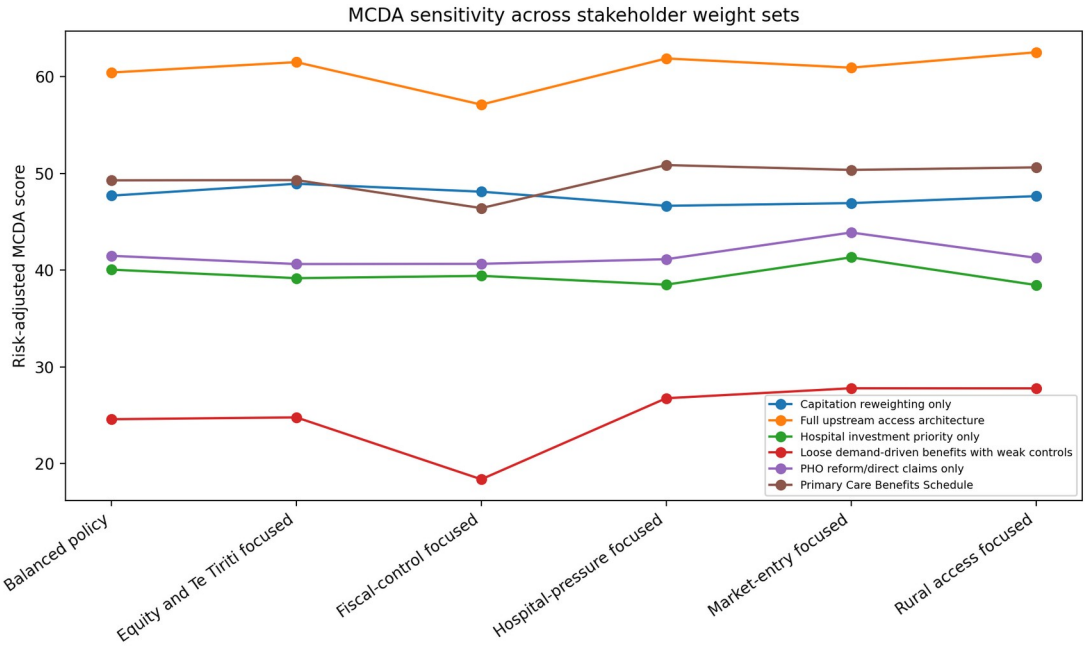
The full upstream access architecture ranks first in the demonstrative MCDA. The Primary Care Benefits Schedule ranks second. Benefits plus scope-enabled eligibility ranks third but is more heavily penalised for implementation and governance risk. Capitation reweighting and capitation reweighting plus an access target perform moderately: they improve allocation and visibility, but do not fully solve marginal supply.

Loose demand-driven benefits with weak controls rank last, despite scoring well on access. This is because the risk-adjustment layer penalises weak equity protections, weak data, weak governance, and high gaming/fiscal exposure.

Sensitivity across stakeholder weight sets

option	Balanced policy	Equity and Te Tiriti focused	Fiscal-control focused	Hospital-pressure focused	Market-entry focused	Rural access focused
ACC/ambulance alternatives strengthened only	6	6	4	4	4	4
Benefits schedule plus scope-enabled eligibility	3	5	6	3	3	2
Capitation reweighting only	5	3	2	6	6	6
Capitation reweighting plus access target	4	4	3	5	4	5
Full upstream access architecture	1	1	1	1	1	1
Hospital investment priority only	8	8	8	8	8	8
Loose demand-driven benefits with weak controls	10	10	10	10	10	10
PHO reform/direct claims only	7	7	7	7	7	7

option	Balanced policy	Equity and Te Tiriti focused	Fiscal-control focused	Hospital-pressure focused	Market-entry focused	Rural access focused
Primary Care Benefits Schedule	2	2	5	2	2	3
Status quo tight control	9	9	9	9	9	9



MCDA sensitivity across stakeholder weight sets

The ranking is not intended to force consensus. It is intended to show where the result is robust and where it depends on stakeholder values.

Interpretation

The MCDA supports a refined version of the policy thesis:

> A demand-driven Primary Care Benefits Schedule may improve marginal supply and hospital-deflection logic, but it becomes a viable policy architecture only when paired with scope governance, equity protections, co-payment calibration, ambulance alternatives, direct/optional claiming, data observability and top-tier KPIs.

This can be expressed as:

> Demand-driven within rules; not demand-driven without rules.

How to use this package

Use the MCDA in three ways:

1. **Internal RACMA discussion**: test whether colleagues accept the game map and priority scores.
2. **Stakeholder workshop**: compare group-specific scores and weight sets.
3. **Empirical validation planning**: turn low-confidence, high-impact criteria into research tasks.

Audit status

This is a structured decision-support artefact. It is not proof of policy effectiveness. It should be updated after OIA responses, stakeholder validation, review findings, fiscal modelling and calibrated simulation.

Included artefacts

- MCDA framework.
- Game-position scoring rubric.
- Policy-option scoring rubric.
- Stakeholder workshop guide.
- Weighting methods note.
- Results interpretation guide.
- Criteria, option, scorecard and weight-set CSVs.
- Blank scoring templates.
- Executable MCDA model.
- Example figures and workbook.